

Feature Engineering

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Transform validation for Feature engineering. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Feature engineering. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the training mean and centered new feature; state units.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

6. Solve the given case

Features are leaf length and width in cm; add length/width.

Compute for 8 and 2; state boundary checks.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A feature is leaf_length divided by leaf_width. A row has width zero. What should the preprocessing rule do before division?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 29 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and leakage test.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Compute features

Compute the worked areas and their training mean.

My reasoning and evidence

13. W2 Transform validation

Center the validation area using the frozen training mean.

My reasoning and evidence

14. W3 Reject leakage

Why is delivery delay invalid for a prediction made before dispatch?

My reasoning and evidence

15. W4 Apply independently

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the training mean and centered new feature; state units.

My reasoning and evidence

16. W5 Plan a fair comparison

How would you test adding area without claiming it must help?

My reasoning and evidence

17. Relationships and confounding — W1 Describe the evidence

State the pattern in the three worked records without using caused or guaranteed.

My reasoning and evidence

18. Relationships and confounding — W2 Identify a confounder

Explain how prior experience could relate to both practice time and score.

My reasoning and evidence

19. Relationships and confounding — W3 Repair an overclaim

A learner predicts every student gains 2 points per extra hour. Explain why that is unsupported.

My reasoning and evidence

20. Relationships and confounding — W4 Plan independently

A fictional dataset links more hints used with lower scores. Give a noncausal explanation and plan a fairer comparison of whether hints help.

My reasoning and evidence

21. Relationships and confounding — W5 State a remaining limit

Even a better comparison has only four fictional groups. What should be reported?

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Feature Engineering

Engineer a feature with a transformation defined before evaluating a new leaf.

Training rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by training mean.
Query length 2, width 4 cm.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Feature Engineering

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Construct area

Query area = $2 \times 4 = 8 \text{ cm}^2$.



2. Fit centering

Training mean = $(2 + 6 + 10) / 3 = 6 \text{ cm}^2$.



3. Apply unchanged transform

Centered query feature = $8 - 6 = 2 \text{ cm}^2$.

CHANGE ONE THING

Include the query in the centering mean.

Explain the changed result using the matching lesson.

What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.